

SBSS Literature Review Comparison (Papers 1-10)

Paper 1

No.: 1

Title: Energy-Efficient Clustered Wireless Sensor Networks for Border Surveillance

Authors: Malathy et al.

Year: 2024

Thought (comparison):

Malathy and colleagues investigate energy-efficient clustered wireless sensor network (WSN) architectures for border surveillance, integrating ant colony optimization (ACO) algorithms for optimal routing performance. Their work addresses the fundamental challenge of balancing computational efficiency, energy consumption, and data security in resource-constrained environments requiring continuous operation. By organizing sensor nodes into hierarchical clusters with ACO-based routing and ECC cryptography, they achieve extended network lifetime and high packet delivery ratios validated through simulation.

In contrast, our proposed work implements direct WiFi bot-to-bot communication with real-time sensing and vision capabilities, avoiding complex multi-hop routing protocols entirely. Unlike their static clustered WSN approach using ACO for network-level routing optimization, we enable mobile, decentralized coordination through autonomous ESP32-CAM units. We emphasize local edge processing on ESP32-CAM for immediate autonomous responses to detected threats, eliminating routing delays and infrastructure complexity. While their system focuses on optimizing data transmission paths across static sensor networks, our design prioritizes on-device intelligence and context-aware decision-making without requiring centralized coordination or complex routing algorithms.

Match Factor: 35%

Paper 2

No.: 2

Title: Integrated Multi-Sensor Data Fusion for Border Control

Authors: Essendorfer et al.

Year: 2024

Thought (comparison):

Essendorfer and colleagues present an integrated framework for border surveillance emphasizing heterogeneous sensor data fusion and transnational information sharing through the SOBCAH project. Their work addresses the complexity of modern border security where infrared cameras, electro-optical systems, radar, and sonar must operate cohesively across vast terrain. The core innovation lies in a unified data fusion architecture integrating disparate sensor streams into a coherent operational picture through probabilistic reasoning, with shared databases enabling real-time information exchange among border agencies across multiple countries.

We emphasize local, real-time decision-making at the edge using ESP32-CAM for immediate autonomous responses, eliminating reliance on centralized databases and cross-border data sharing protocols used in their centralized architecture. Unlike their approach requiring complex multi-sensor fusion infrastructure and transnational coordination, our system performs human-animal classification directly on-device, enabling independent operation without external server dependencies. While their work excels at large-scale interoperability and multi-agency coordination, our design prioritizes autonomous edge intelligence, privacy preservation through local processing, and simplified deployment suitable for remote border regions lacking complex infrastructure. Our ESP32-CAM units can operate completely offline, making decisions locally without requiring the extensive communication infrastructure their system depends upon.

Match Factor: 30%

Paper 3

No.: 3

Title: Swarming Control for Unmanned Vehicles in Surveillance

Authors: Sauter et al.

Year: 2024

Thought (comparison):

Sauter and colleagues demonstrate swarming control algorithms for unmanned ground and aerial vehicles applied to surveillance and infrastructure protection. Their digital pheromone-based approach enables autonomous, adaptive coordination of heterogeneous sensor platforms, supporting complex behaviors including target tracking and multi-sensor fusion. The system mimics biological swarm intelligence where individual agents follow simple rules to achieve sophisticated collective behavior without centralized control, demonstrating robustness against individual agent failures and scalability to large deployments through real-world tests with UAVs, UGVs, and sensor networks.

We simplify swarm coordination into a practical, low-cost dual-bot system using direct WiFi communication, suitable for embedded deployment without the computational complexity of their digital-pheromone-based approach. Unlike their sophisticated swarm algorithms requiring complex coordination mechanisms and substantial computational resources, our ESP32-CAM based system implements straightforward bot-to-bot communication protocols optimized for resource-constrained edge devices. While their work demonstrates advanced multi-agent coordination with digital pheromones and complex behavioral algorithms, we prioritize deployment simplicity, cost-effectiveness, and immediate practicality for border surveillance. Our approach

trades some coordination sophistication for dramatically reduced system complexity, lower unit costs, and easier field deployment, making it more accessible for resource-limited border security applications.

Match Factor: 40%

Paper 4

No.: 4

Title: IoT-Based Smart Border Security with Machine Learning

Authors: Fatima et al.

Year: 2024

Thought (comparison):

Fatima and colleagues propose an IoT-based smart border security system utilizing machine learning for intruder detection, employing sensors, thermal imaging, and camera data analyzed by Faster R-CNN and EfficientDet models. Their architecture integrates edge computing for real-time processing with cloud-encrypted storage for long-term data retention and analysis, with automatic SMS alerts notifying security personnel upon threat detection. The system addresses practical deployment challenges in rugged terrains with limited manpower through multi-stage verification to mitigate false alarms and cloud integration for scalable remote monitoring.

We run lightweight CNN directly on ESP32-CAM through TinyML for offline operation with low latency, unlike their cloud-based ML models requiring constant connectivity and thermal imaging infrastructure. While their system depends on cloud resources for running computationally intensive models like Faster R-CNN and EfficientDet, our approach implements a custom lightweight neural network optimized specifically for ESP32-CAM's limited resources, enabling complete autonomous operation without internet connectivity. Unlike their requirement for thermal imaging cameras and cloud infrastructure, we achieve human-animal differentiation using standard RGB cameras with on-device inference, drastically reducing deployment costs and infrastructure requirements. Our system operates effectively in remote areas without cellular coverage or internet access, addressing a critical limitation in their cloud-dependent architecture.

Match Factor: 55%

Paper 5

No.: 5

Title: Wireless Sensor Networks for Border Surveillance

Authors: Arjun et al.

Year: 2024

Thought (comparison):

Arjun and colleagues survey multiple wireless sensor network (WSN) techniques for border surveillance and intruder detection, highlighting hybrid systems integrating seismic, acoustic, infrared, and camera sensors to detect human intrusions in diverse border scenarios. The paper discusses multi-layered architectures and communication protocols critical for large-scale deployments, examining how different sensor modalities complement each other through multi-sensor fusion approaches. They analyze challenges including false alarms from wildlife, sensor coverage limitations, and energy constraints, proposing intelligent data aggregation and energy-efficient communication protocols to extend network lifetime.

We move beyond theoretical discussion by building and validating a working multi-sensor (PIR, ultrasonic) and vision-based prototype demonstrating practical implementation challenges and solutions. Unlike their comprehensive survey which primarily discusses theoretical architectures and proposed approaches, our work provides a functional, tested system addressing real-world deployment constraints. While they examine various sensor fusion methodologies abstractly, we implement practical sensor integration on resource-constrained ESP32-CAM hardware, demonstrating how lightweight CNN models can effectively process camera data alongside PIR and ultrasonic sensors for human-animal differentiation. Our contribution includes actual hardware implementation details, power consumption measurements, and field-tested performance metrics rather than simulation results or theoretical frameworks, filling the gap between their surveyed concepts and deployable border surveillance solutions.

Match Factor: 45%

Paper 6

No.: 6

Title: Machine Learning Approaches Survey

Authors: Thomas et al.

Year: 2024

Thought (comparison):

Thomas and colleagues present a comprehensive survey on machine learning approaches, categorizing them into supervised, unsupervised, semi-supervised learning, and reinforcement learning methods. The paper explains key algorithms including decision trees, rule-based classifiers, naive Bayes, k-nearest neighbors, neural networks, and support vector machines, providing foundational understanding of their operational principles. They discuss applications across domains like pattern recognition, natural language processing, computer vision, and autonomous systems, offering insights directly relevant for AI-driven border surveillance and detection systems.

We apply ML concepts specifically by implementing CNN-based TinyML model on resource-constrained ESP32-CAM hardware for real-time visual intrusion detection, moving from theory to application. Unlike their broad theoretical survey covering multiple machine learning paradigms, our work focuses on practical deployment of convolutional neural networks optimized for edge devices with severe computational and

memory constraints. While they discuss various algorithms academically, we address the engineering challenges of quantizing, optimizing, and deploying neural networks on microcontrollers with less than 8MB RAM. Our contribution demonstrates how theoretical ML concepts translate into functioning edge-AI surveillance systems, including model compression techniques, inference optimization strategies, and real-time performance under resource constraints—practical aspects their survey does not explore.

Match Factor: 30%

Paper 7

No.: 7

Title: YOLO-Based Real-Time Border Security

Authors: Nahata et al.

Year: 2024

Thought (comparison):

Nahata and colleagues present a YOLO-based real-time border security surveillance system integrating deep learning with camera and UAV inputs to detect humans, vehicles, drones, and weapons. The paper highlights challenges in traditional border monitoring including limited visibility, vast coverage areas, and manual patrol inefficiencies. Their automated object detection approach using YOLO (You Only Look Once) architecture enables timely threat recognition with improved detection speed and accuracy, supporting integration with IoT devices for extended coverage and providing integrated alert mechanisms for rapid notification.

We replace YOLO with custom lightweight CNN optimized for ESP32-CAM's limited resources, trading minimal accuracy for drastically lower power consumption and cost, enabling edge processing. Unlike their approach requiring substantial computational resources to run YOLO models (typically on GPUs or high-end embedded platforms like Jetson boards), our lightweight CNN operates directly on ESP32-CAM's modest microcontroller. While YOLO-based systems excel in detection accuracy and multi-class object recognition, they require continuous power supply and expensive hardware unsuitable for distributed, solar-powered border deployments. Our design prioritizes energy efficiency, extended autonomous operation, and ultra-low unit cost (under \$10 per node) over detection sophistication, making large-scale deployment economically feasible. This trade-off enables deploying hundreds of units for comprehensive coverage at the cost of a few YOLO-based systems.

Match Factor: 60%

Paper 8

No.: 8

Title: Autonomous Robot Swarm for Border Intrusion

Authors: Mahjoub et al.

Year: 2024

Thought (comparison):

Mahjoub and colleagues develop an autonomous wireless sensor network employing a swarm of mobile robots for border intrusion detection, mimicking natural swarm behavior through distributed coordination algorithms. Their heterogeneous robot system collaboratively navigates complex terrains using YOLOv5 for object identification, with ROS (Robot Operating System) and Gazebo simulation environment utilized for system design, testing, and coordination protocol development. The study emphasizes cooperative navigation strategies, robust multi-robot communication protocols, and fault tolerance mechanisms enabling continued operation when subset of robots malfunction.

We adopt simpler, non-ROS architecture with low-power WiFi communication, prioritizing ease of deployment and cost-effectiveness over complex swarm behaviors using YOLOv5 and ROS. Unlike their sophisticated robotic platform requiring ROS middleware, complex simulation environments, and computationally intensive YOLOv5 models, our ESP32-CAM based system operates with minimal software dependencies and straightforward programming. While their approach demonstrates advanced multi-robot coordination and simulation-validated behaviors, it requires significant technical expertise for deployment and maintenance. Our design eliminates ROS complexity, replacing YOLOv5 with lightweight CNN suitable for microcontroller execution, and implements simple yet effective WiFi-based coordination that non-specialist personnel can deploy and maintain. This architectural simplification reduces development time, lowers technical barriers, and enables rapid field deployment without extensive simulation or complex coordination protocols.

Match Factor: 50%

Paper 9

No.: 9

Title: Mini Seeker Robot for Border Surveillance

Authors: Reddy et al.

Year: 2024

Thought (comparison):

Reddy and colleagues propose the Mini Seeker Robot, an automated border surveillance platform equipped with ultrasonic sensors, facial recognition, and live video streaming functionalities. Controlled via Raspberry Pi microcontroller and integrated with OpenCV computer vision library, the robot detects obstacles autonomously and identifies individuals by matching facial data with pre-registered databases. The cost-effective solution operates autonomously following pre-programmed patrol routes or under remote manual control, with ultrasonic sensors enabling collision avoidance and facial recognition providing identity verification capabilities.

We substitute Raspberry Pi with ESP32-CAM paired with lightweight ML, achieving significant reduction in unit cost, power draw, and physical footprint for wider field deployment. Unlike their Raspberry Pi-based platform which consumes approximately 2.5-4W during operation and costs \$35-75 per unit (depending on model), our ESP32-CAM system operates at under 1W and costs approximately \$8-12 per unit. While Raspberry Pi provides greater computational flexibility for running OpenCV and complex facial recognition algorithms, it requires substantially more power, making solar-powered deployment challenging. Our approach trades facial recognition capability for human-animal classification—a more relevant distinction for border surveillance—while achieving dramatically longer battery life and lower deployment costs. This enables deploying 5-10 ESP32-CAM units for every Raspberry Pi robot, providing superior coverage density and system redundancy.

Match Factor: 65%

Paper 10

No.: 10

Title: Helmet Detection and Number Plate Recognition

Authors: Sathe et al.

Year: 2024

Thought (comparison):

Sathe and colleagues focus on helmet detection and number plate recognition using deep learning models YOLOv5 and EasyOCR within traffic surveillance contexts. Their methodology involves bounding box overlap techniques to associate detected helmets with corresponding motorcyclists, effectively identifying helmet violation instances for traffic law enforcement. The study achieves high accuracy through transfer learning, leveraging pre-trained models and fine-tuning on domain-specific datasets, with comprehensive dataset augmentation including rotation, scaling, brightness adjustment, and synthetic variations improving model robustness.

We repurpose real-time object classification technology for identifying security threats (humans vs. animals) in border surveillance, adapting traffic monitoring techniques to security applications. Unlike their focus on traffic law enforcement through helmet and license plate detection, our system applies similar deep learning principles to border security contexts requiring human-animal differentiation. While their work demonstrates effective use of YOLOv5 and transfer learning for specific traffic scenarios, we adapt these methodologies to embedded edge devices with severe resource constraints. Their approach requires substantial computational resources for real-time video processing with YOLOv5 and OCR, whereas we implement simplified classification models optimized for ESP32-CAM's limited memory and processing capabilities. Our contribution demonstrates how traffic surveillance computer vision techniques can be efficiently adapted and compressed for deployment on ultra-low-cost edge devices in remote border monitoring applications.

Match Factor: 35%

Status: Papers 1-10 completed **Remaining:** Papers 11-50 **Ready for:** Color coding (to be applied later)